

Inban Tamizhselvan

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EDUCATION

University of Massachusetts Dartmouth

Jan 2023 - Aug 2024

Master of Science in Computer and Information Science

Rajalakshmi Engineering College, Tamil Nadu, India

Aug 2018 - May 2022

Bachelor of Engineering in Computer Science

ACADEMIC PROJECTS

Ishopy E-commerce Website

Aug 2024 – Oct 2024

- Constructed a scalable e-commerce platform using **React.js**, **Node.js**, **MongoDB**, and **Context API** for efficient state management.
- Enforced **JWT authentication** for secure user logins and product management, improving security compliance by **50%**.
- Built a responsive UI using **TailwindCSS**, optimizing mobile responsiveness and **user experience**.
- Integrated a **shopping cart system** featuring real-time price updates, elevating user experience incorporating **email subscription** functionalities through **Nodemailer** for promotional offers and utilizing **Stripe** for secure payment processing.
- Tested APIs comprehensively with **Postman** and utilized **Git/GitHub** for version control and team collaboration.

Real-time Serverless Chat Application

Aug 2024 - Sep 2024

University of Massachusetts Dartmouth

- Developed a real-time chat application using **React.js**, **CSS3**, and **JavaScript**, integrating **Firebase** for authentication, real-time data synchronization, and cloud storage, ensuring a responsive **user experience** with **<100ms latency**.
- Utilized **Firebase Authentication** for secure user login and registration, incorporating **WebRTC** for peer-to-peer communication, allowing real-time audio and video streaming between users without the need for additional servers.
- Leveraged **Zustand** for **state management**, optimizing asynchronous operations and state handling to ensure an uninterrupted, lag-free chat experience with real-time updates.

Decentralized NFT Marketplace and Token Transfer using ICP Blockchain

Jun 2024 - Jul 2024

University of Massachusetts Dartmouth

- Engineered a **decentralized token system** allowing users to authenticate via **ICP Internet Identity**, claim tokens through a faucet, perform token transfers, and maintain a detailed transaction history, enhancing platform security and user trust.
- Implemented **Motoko-based smart contracts** to create NFT marketplace where users can mint, buy, and sell NFTs, leveraging inter-canister communication between NFT transactions and transfer system, ensuring **100%** transparency.
- Improved frontend using **React.js**, **React Router**, and **CSS3**, increasing user engagement by **35%**.
- Managed development using **Git version control**, ensuring smooth deployment and scalability.

To-Do List Application

May 2024 - Jun 2024

University of Massachusetts Dartmouth

- Created a **To-Do List** web application with **JavaScript**, **Embedded JavaScript (EJS)**, **Express.js**, **HTML**, and **CSS**, ensuring robust integration between frontend and backend components for a seamless experience.
- Enhanced **data management** by implementing **PostgreSQL**, achieving a **40%** increase in scalability and performance.
- Designed and deployed **RESTful APIs** to enable smooth and secure communication between the **client-side** and **server-side** systems, facilitating real-time data interactions and speeding up system responsiveness.
- Applied **Material-UI** and **React.js** for an intuitive interface, improving the overall experience by **30%**.

SKILLS

- **Frontend:** HTML, CSS, Bootstrap, JavaScript (ES6+), React, Zustand, Redux (state management), jQuery, TailwindCSS.
- **Backend:** Node.js, Express.js, EJS, RESTful APIs, Stripe, GraphQL, Motoko, Cloud Firestore, NPM, WebRTC.
- **Databases:** PostgreSQL, MongoDB (NoSQL), MySQL (RDBMS), Firebase Database.
- **Programming Languages:** Python (Django), Java, C, C++, PHP.
- **Tools:** Git and GitHub, Command Line Interface (CLI), Visual Studio Code, Microsoft Office Suite, AWS, Postman.
- **Additional Skills:** Verbal Communication, Teamwork, Software Development Lifecycle (SDLC).